

RubricEye Exam

Answer Booklet

Name: _____

Candidate ID: _____

Date: _____ Exam Version: RE-01

Answer only the questions you choose.

Keep all writing and drawings inside the bordered areas.
For diagram/flowchart responses, use clear labels and arrows.

Q2(i)

4 marks

Newton's Second law of motion states that the acceleration of an object is directly proportional to the net force acting on it & inversely proportional to its mass.

The direction of acceleration is the same as the direction of the net force.

Mathematical expression:

$$F = ma$$

Where,

F = Net force m = Mass a = Acceleration

Q2(ii)

4 marks

	RAM	ROM
Full form	Random Access Memory	Read only Memory
Function	Temporary memory Data is lost when power is off	Persistent memory Data is not lost when power is off
Read/write	Can be read & written	Mostly can only be read
Example	8GB DDR4 Ram in a Computer	BIOS Chip / Firmware in a computer

RubricEye Exam

Q2(iii)

4 marks

The nucleus is called the control center of the cell. Roles:

1. contain DNA which carries genetic information
2. controls all cell activities like growth, metabolism, & reproduction-
3. controls protein synthesis by sending instructions through RNA.
4. protects the genetic material of the cell.

Q2(iv)

4 marks

Q2(v)

4 marks

Q2(vi)

4 marks

For a box resting & being pushed on a horizontal surface, the 4 forces are:

Upward: Normal force / Reaction force N

- The surface pushes up on the box.

Downward: weight / force due to gravity $W = mg$

- Earth pulls the box down.

Forward: applied force F

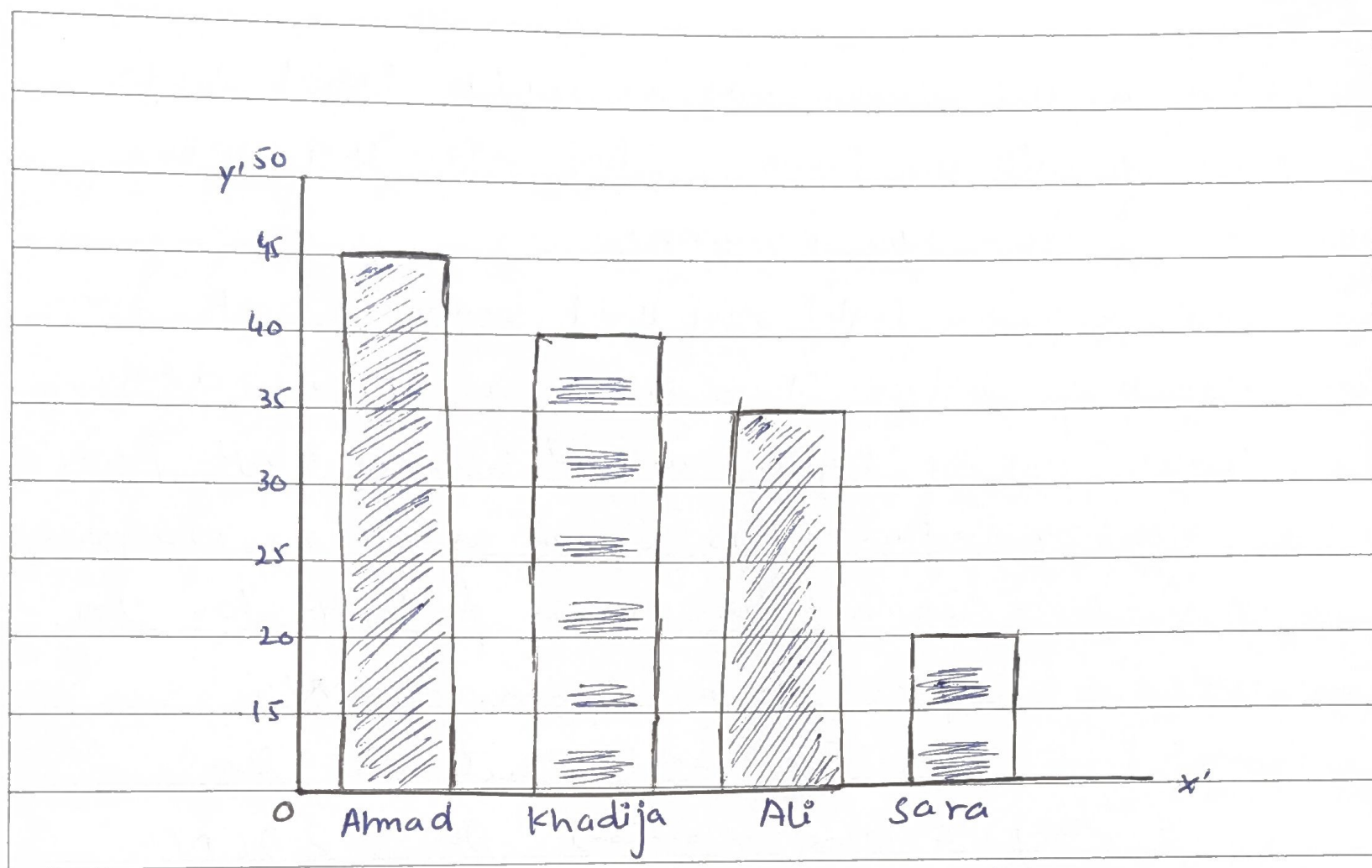
- The push/pull force acting in the direction of motion

Backward: frictional force f

- The surface opposes the motion.

Q2(vii)

4 marks



Observations: Ahmad got the highest marks & Sara got the lowest marks.

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Q3(A)

8 marks

Reflection is the phenomenon in which light rays bounce back into the same medium when they strike a smooth surface like a mirror.

- 1- Incident Ray: The light ray that bounces back from the reflecting surface. It is shown as AB in diagram.
 - 2- Reflected Ray: The light ray that bounces back from the reflecting surface. It is shown as OB in diagram.
 - 3- Normal: An imaginary line drawn perpendicular to the reflecting surface. It is shown as ON.
 - 4- Angle of incidence: The angle between the incident ray & the normal. It is $\angle AON$.
 - 5- Angle of Reflection: The angle between the reflected ray & the normal. It is $\angle BON$.
- Law of Reflection:

Law 1: The incident ray, the reflected ray, & the normal at the point of incidence all lie in the same plane. This means all three are on the same paper/surface when we draw them.

Law 2: The angle of incidence is always equal to the angle of reflection.

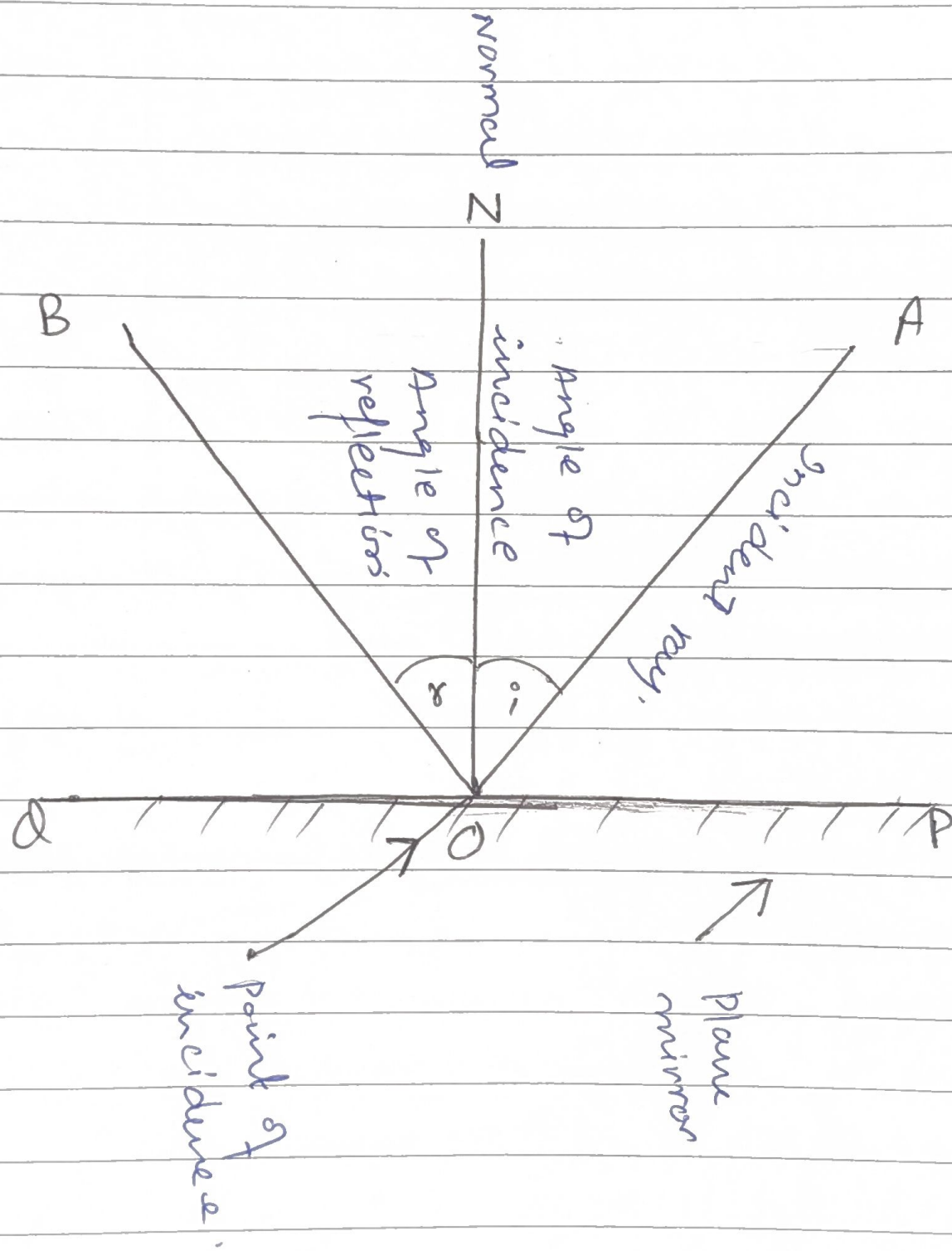
Mathematically: $\angle i = \angle r$.

So, the relationship is: Angle of incidence = Angle of reflection

Q3(B)

7 marks

According to law of Reflection.



8 marks

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

7 marks

ROUGH WORK

A large, empty rectangular box with a thin black border, occupying the central portion of the page. It is intended for students to show their rough work or calculations during an exam.